Certificate of Analysis

NRC·CNRC Certified Reference Material

FEBS-1

Otolith Certified Reference Material for Trace Metals

FEBS-1 is a saggital otolith reference material procured from red snapper (*Lutjanus campechanus*). Certified values are based on unweighted mean results from data submitted by collaborating laboratories. The expanded uncertainty (U_{CRM}) in the certified value is equal to $U = ku_c$ where u_c is the combined standard uncertainty calculated according to the JCGM Guide [1] and k is the coverage factor. The value of u_c is determined from the combined uncertainties of the various analytical methods (u_{char}) as well as uncertainties associated with homogeneity (u_{hom}). It is intended that U_{CRM} encompasses every aspect that reasonably contributes to the uncertainty of the measurement [2]. A coverage factor k=2 was applied for all elements. The table below lists those elements for which certified values have been established.

Table 1: Certified quantity values for FEBS-1

Element	Mass fraction, mg/kg
barium (d,i,p)	5.09 ± 0.23
lithium (d,p)	0.305 ± 0.044
magnesium (d,i,p)	23.6 ± 1.3
manganese (d,n)	0.686 ± 0.016
sodium (d,i,n)	2594 ± 161
strontium (d,i,p)	2055 ± 79

Table 1 (continued): Certified quantity values for FEBS-1

Element	Mass fraction, g/100 g
calcium (d,i,n,x)	38.3 ± 1.4

Coding

The coding refers only to the instrumental method of analyte determination.

- **d** Inductively coupled plasma mass spectrometry (ICPMS)
- **i** Inductively coupled plasma atomic emission spectrometry (ICP-AES)
- **n** Neutron activation analysis (INAA)
- **p** Isotope dilution ICPMS (ID-ICPMS)
- **x** x-ray fluorescence spectrometry (XRF)

Intended Use

This reference material is primarily intended for use in the calibration of procedures and the development of methods used for the analysis of bulk elemental composition of otoliths and other marine aragonites.

Storage and sampling

This material should be kept tightly closed in the original bottle and should be stored in a cool location. The contents should be well mixed by rotation and shaking of the bottle prior to use. The bottle should be tightly closed immediately after use.

Instructions for drying

FEBS-1 can be dried to constant mass in an oven for 3 hours at 105 ^{o}C.

Preparation of FEBS-1

The otoliths were extracted from the fish, rinsed in ultrapure water, air-dried, and stored in polyethylene bottles. Surface contamination was later removed in a class 100 clean room by rest-rinsing all otoliths in 2% ultrapure HNO_3 for 5 minutes, followed by triple-rinsing in ultrapure water. Otoliths were subsequently sonicated in ultrapure water for 10 minutes, triple-rinsed again in ultrapure water and allowed to dry under a laminar flow hood for 72 hours. The otoliths were crushed using a ball mill and the resulting powder was sieved through 100 and 200 mesh stainless steel sieves (150 µm and 75 µm particle sizes, respectively). The 100+ and 200+ fractions were returned to the mill and ground until all material passed the 200 mesh sieve. The material was then radiation sterilized at a minimum dose of 25 kGy (Neutron Products Inc., Dickerson, MD, USA).

Stability

As a result of interest in only the total amount content of the major, minor and trace elements, coupled with the chemical and physical properties of the matrix, the need for any rigorous stability testing of this material under its recommended conditions of storage was obviated and, as such, contributions to a specific long-term stability component of the uncertainty were not included in the evaluation of the combined uncertainty presented in Table 2.

Certified value

The results from a sub-group of participants were used for the certification of FEBS-1. These laboratories were selected based on their previous experience analysing otolith samples. The certified values were calculated from the unweighted means of the results of the participating laboratories [3,4].

Guidelines for CRM producers suggest all sources relevant to the user of the material should contribute to the uncertainty of the certified value [2-4]. Included in the overall uncertainty estimate are uncertainties in the batch characterization (u_{char}), uncertainties related to possible between-bottle variation (u_{hom}) as well as instability derived from effects relating to long-term storage and transport (u_{stab}). The various components used to calculate the certified values are shown in Table 2.

Characterization

The characterization uncertainties (u_{char}) were calculated from the standard deviation of the means [4].

Homogeneity

The material was tested for homogeneity at NRC using ICP-MS or ICP-OES. Results from triplicate sub-samples (250 mg) from at least 8 bottles were evaluated using ANOVA [5]. Table 2 summarizes the resulting uncertainty components for homogeneity for each element at the 250 mg sub-sample size used for the testing.

Table 2: Uncertainty components for FEBS-1

Uncertainty	$u(w_{Ba})$, mg/kg	$u(w_{Ca})$, g/100g	$u(w_{Li})$, mg/kg	$u(w_{Mg})$, mg/kg	$u(w_{Mn})$, mg/kg	$u(w_{Na})$, mg/kg	$u(w_{Sr})$, mg/kg
u_{char}	0.039	0.62	0.0211	0.395	0.006	72.9	21
u_{hom}	0.108	0.35	0.0055	0.535	0.0055	23.5	34
u_c	0.115	0.717	0.022	0.665	0.0081	80.3	40
$U(k=2)$	0.23	1.4	0.044	1.33	0.016	161	79

Information values

Significant heterogeneity was observed for many elements, precluding their certification. A mass fraction interval is presented in the following table, which simply summarizes both the highest and lowest determined of the element in any given subsample by the primary IDICPMS technique along with corroboration by another methodology. These data may be of interest to users/readers and hence are included in this report. Table 3 shows the interval of the extreme values of mass fractions obtained during sample characterization.

Table 3: Information values for FEBS-1

Element	Interval	Technique
cadmium	1.4 - 3.2 µg/kg	IDICPMS, ICPMS
copper	4.2 - 6.8 mg/kg	IDICPMS, ICPMS
nickel	16 - 29 µg/kg	IDICPMS
lead	0.40 - 0.77 mg/kg	IDICPMS, ICPMS
zinc	3.2 - 6.3 mg/kg	IDICPMS, INAA

Metrological traceability

Results presented in this certificate are traceable to the SI through gravimetrically prepared standards of established purity and international measurement intercomparisons. As such, they serve as suitable reference materials for laboratory quality assurance programs, as outlined in ISO/IEC 17025. This CRM is registered at the Bureau International des Poids et Mesures (BIPM) in Appendix C of the Comité International des Poids et Mesures database listing Calibration and Measurement Capabilities accepted by signatories to the Mutual Recognition Arrangement of the Metre Convention.

Quality Management System (ISO 17034, ISO/IEC 17025)

This material was produced in compliance with the NRC Metrology Quality Management System, which conforms to the requirements of ISO 17034 and ISO/IEC 17025. The Metrology Quality Management System supporting the NRC Calibration and Measurement Capabilities, as listed in the *Bureau international des poids et mesures* (BIPM) Key Comparison Database (http://kcdb.bipm.org/), has been reviewed and approved under the authority of the Inter-American Metrology System (SIM) and found to be in compliance with the expectations of the *Comité international des poids et mesures* (CIPM) Mutual Recognition Arrangement. The SIM approval is available upon request.

Updates

Users should ensure that the certificate they have is current. Our web site at www.nrc.gc.ca/crm will contain any new information.

References

1. Evaluation of measurement data – Guide to the expression of uncertainty in measurement JCGM 100:2008.

2. J. Pauwels, A. van der Veen, A. Lamberty, H. Schimmel, Evaluation of uncertainty of reference materials. Accred Qual Assur (2000) 5:95-99.

3. A. M.H.van der Veen and J. Pauwels, Uncertainty calculations in the certification of reference materials. 1.Principles of analysis of variance. Accred Qual Assur (2000) 5:464–469.

4. A. M.H. van der Veen, T. P.J. Linsinger, H. Schimmel, A. Lamberty and J. Pauwels, Uncertainty calculations in the certification of reference materials 4. Characterisation and certification. Accred Qual Assur (2001) 6:290–294.

5. T.P.J. Linsinger, J. Pauwels, A.M.H. van der Veen, H. Schimmel and A. Lamberty, Accred. Qual. Assur., 2001, 6, 20 – 25.

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Z. Chen, Laboratory for Isotope and Trace Element Research Department of Chemistry and Biochemistry Old Dominion University, Norfolk, VA 23529 USA

S. Thorrold, Biology Department, Woods Hole Oceanographic Institution, Woods Hole, MA 02543, USA

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Approved by:

Zoltan Mester, Ph. D.
Team Leader, Chemical Metrology
NRC Metrology

This Certificate is only valid if the corresponding product was obtained directly from NRC or one of our qualified vendors.

Comments, information and inquiries should be addressed to:

National Research Council Canada
Métrologie
1200 Montreal Road, Building M-12
Ottawa, Ontario K1A 0R6

Telephone: 1-613-993-2359
Fax: 1-613-993-2451
Email: CRM-MRCInorganic-Inorganiques@nrc-cnrc.gc.ca

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